第二章 碳和硅

Chapter 2 Carbon and Silicon

第一节 碳

Section 1 Carbon

一、碳的性质

Ⅰ. Properties of Carbon

在常温下，碳(carbon，C)的化学性质不活泼；但是，在高温下，碳能与许多物质发生反应。

At room temperature, carbon (C) is chemically inert. However, at high temperatures, carbon can react with many substances.

1. 碳与氧气的反应

1. Reaction of Carbon with Oxygen

碳在氧气或空气中充分燃烧时，生成二氧化碳(CO_2)，同时放出大量的热。

When carbon burns sufficiently in oxygen or air, carbon dioxide (CO_2) is produced, and a large amount of heat is released at the same time.

$$C + O_2 \overset{\text{点燃/Ignition}}{=\!=} CO_2$$

碳燃烧不充分时, 生成一氧化碳 (CO), 同时放出热量。

When carbon burns incompletely, it forms carbon monoxide (CO), also releasing heat.

$$2C + O_2 \overset{\text{点燃/Ignition}}{=\!=} 2CO$$

【例 1】碳在氧气中充分燃烧，生成____；碳在氧气中不充分燃烧，生成____。

【Example 1】When carbon burns completely in oxygen, it forms____; when carbon burns incompletely in oxygen, it forms____.

解题思路:要解答该问题，需先明确碳与氧气反应的关键影响因素——氧气的充足程度。当氧气供应充足时，碳能发生完全燃烧反应，此时产物为二氧化碳，且伴随大量放热；当氧气量不足时，碳无法完全燃烧，会生成一氧化碳，同样会释放热量。结合碳燃烧的两种反应规律，即可得出对应的产物答案。

Solution idea: To answer this question, it is necessary to first clarify the key influencing factor in the reaction of carbon with oxygen - the sufficiency of oxygen. When oxygen is sufficient, carbon can undergo a complete combustion reaction, and the product is carbon dioxide, accompanied by a large amount of heat release; when the amount of oxygen is insufficient, carbon cannot burn completely and carbon monoxide will be formed, also releasing heat. Combining the two reaction laws of carbon combustion, the corresponding product answers can be obtained.

2. 碳与某些氧化物的反应

2. Reaction of Carbon with Certain Oxides

单质碳具有还原性。例如, 在加热条件下, 木炭能够将黑色的氧化铜还原为红色的金属铜, 同时放出二氧化碳气体，如图 2.1-1 所示。

Elemental carbon has reducing properties. For example, under heating conditions, charcoal can reduce black copper oxide to red metallic copper, while releasing carbon dioxide gas, as shown in Figure 2.1-1.

$$2CuO + C \xlongequal{\text{高温/High temperature}} 2Cu + CO_2 \uparrow$$

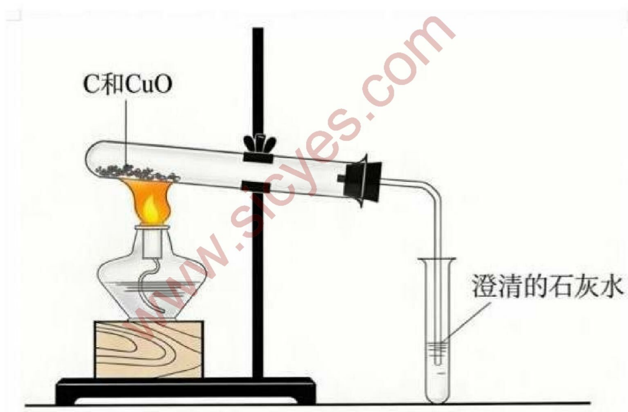

图 2.1-1 木炭还原氧化铜

Figure 2.1-1 Reduction of Copper Oxide by Charcoal

热的碳还能将二氧化碳还原为一氧化碳。

Hot carbon can also reduce carbon dioxide to carbon monoxide.

$$CO_2 + C \xlongequal{\text{高温/High temperature}} 2CO$$

二、二氧化碳的性质

Ⅱ. Properties of Carbon Dioxide

二氧化碳是一种无色、无味的气体。标准状况下，二氧化碳的密度是空气的 1.5 倍。二氧化碳能溶于水。

Carbon dioxide is a colorless and odorless gas. Under standard conditions, the density of carbon dioxide is 1.5 times that of air. Carbon dioxide is soluble in water.

固体二氧化碳称为干冰。在压强为 101 kPa、温度为 −78.5°C 时，干冰能够升华，直接变成二氧化碳气体。

Solid carbon dioxide is called dry ice. At a pressure of 101 kPa and a temperature of −78.5°C , dry ice can sublime and directly turn into carbon dioxide gas.

空气中约含有 0.03% 的二氧化碳，但受人类活动 (如化石燃料燃烧) 影响，近年来空气中的二氧化碳含量猛增，导致温室效应(greenhouse effect)加剧，全球气候变暖。

The air contains about 0.03% carbon dioxide. However, affected by human activities (such as the burning of fossil fuels), the carbon dioxide content in the air has increased sharply in recent years, leading to an intensified greenhouse effect and global warming.

1. 二氧化碳与水的反应

1. Reaction of Carbon Dioxide with Water

二氧化碳是酸性氧化物，与水反应生成碳酸(carbonic acid, H_2CO_3)。碳酸能够使紫色 (purple) 的石蕊 (litmus) 试液变成红色。

Carbon dioxide is an acidic oxide and reacts with water to form carbonic acid (H_2CO_3). Carbonic acid can turn purple litmus solution red.

$$CO_2 + H_2O = H_2CO_3$$

碳酸很不稳定，很容易分解生成二氧化碳和水。

Carbonic acid is very unstable and easily decomposes into carbon dioxide and water.

$$H_2CO_3 = CO_2\uparrow + H_2O$$

2. 二氧化碳的鉴别 (differentiate)

2. Identification (differentiate) of Carbon Dioxide

当向澄清的石灰水[①](limewater)里通入二氧化碳时，石灰水变混浊(muddy)，这是因为生成了白色的碳酸钙 ($CaCO_3$) 沉淀(precipitate)。

When carbon dioxide is passed into clear limewater [①] , the limewater becomes turbid because a white calcium carbonate ($CaCO_3$) precipitate is formed.

$$CO_2 + Ca(OH)_2 = CaCO_3\downarrow + H_2O$$

[①]主要成分为 $Ca(OH)_2$ 的水溶液。[①]mainly consists of an aqueous solution of $Ca(OH)_2$.

我们可以利用这个反应来鉴别二氧化碳。

We can use this reaction to identify carbon dioxide.

二氧化碳一般不燃烧，也不支持燃烧，常被用作灭火剂。

Carbon dioxide generally does not burn and does not support combustion, and is often used as a fire extinguisher.

【例 2】向澄清的石灰水里通入二氧化碳，石灰水变混浊，这是因为生成了白色的(　)

【Example 2】When carbon dioxide is passed into clear lime water, the limewater becomes turbid because a white (　) is formed

A. $BaCO_3$　B. $CaCO_3$　C. $Ca(OH)_2$　D. CaO

解题思路: 首先明确澄清石灰水的主要成分是 $Ca(OH)_2$,通入 CO_2 后发生化学反应。根据化学反应前后元素种类不变的原则，反应物中含有的元素为 Ca 、 O 、 H 、 C ，选项 A 中含 Ba 元素，反应物中无 Ba 元素，可排除 A；选项 C 是澄清石灰水的主要成分，是反应物而非反应生成的物质，排除 C；选项 D 为氧化钙，该反应的反应条件和反应物均无法生成氧化钙，排除 D；而 CO_2 与 $Ca(OH)_2$ 反应会生成不溶于水的白色 $CaCO_3$ 沉淀，导致石灰水变浑浊，因此正确答案为 B。

Problem-solving idea: First, clarify that the main component of clear lime water is $Ca(OH)_2$. After passing in CO_2 , a chemical reaction occurs. According to the principle that the types of elements remain unchanged before and after a chemical reaction, the elements contained in the reactants are Ca 、 O 、 H 、 C . Option A contains the Ba element, and there is no Ba element in the reactants, so A can be excluded; Option C is the main component of clear lime water and is a reactant rather than a product of the reaction, so C is excluded; Option D is calcium oxide, and the reaction conditions and reactants of this reaction cannot produce calcium oxide, so D is excluded; while CO_2 reacts with $Ca(OH)_2$ to form an insoluble white $CaCO_3$ precipitate, causing the lime water to become turbid. Therefore, the correct answer is B.

3. 二氧化碳的实验室制法

3. Laboratory Preparation of Carbon Dioxide

在实验室里, 二氧化碳常用稀盐酸与碳酸钙反应来制取, 其化学反应方程式为:

In the laboratory, carbon dioxide is commonly prepared by reacting dilute hydrochloric acid with calcium carbonate. The chemical reaction equation is:

$$CaCO_3 + 2HCl = CaCl_2 + CO_2 \uparrow + H_2O$$

制取二氧化碳的装置如图 2.1-2 所示。

The device for preparing carbon dioxide is shown in Figure 2.1-2.

图 2.1-2 制取二氧化碳的装置

Figure 2.1-2 Device for Preparing Carbon Dioxide

由于二氧化碳能溶于水，因此不能用排水法收集；利用二氧化碳的密度大于空气这一性质，我们可以采用向上排空气法来收集二氧化碳。

Since carbon dioxide is soluble in water, it cannot be collected by the water displacement method. Taking advantage of the property that carbon dioxide is denser than air, we can use the upward displacement of air method to collect carbon dioxide.

三、一氧化碳的性质

III. Properties of Carbon Monoxide

一氧化碳是无色、无味的气体，通常状况下，比空气略轻，难溶于水。

Carbon monoxide is a colorless and odorless gas. Under normal conditions, it is slightly lighter than air and is sparingly soluble in water.

1. 可燃性

1. Combustibility

一氧化碳在空气里燃烧，火焰呈蓝色，生成二氧化碳，同时放出大量的热。

Carbon monoxide burns in air with a blue flame, producing carbon dioxide and releasing a large amount of heat.

$$2CO + O_2 \overset{\text{点燃/Ignition}}{=\!=} 2CO_2$$

2. 还原性

2. Reducing Property

一氧化碳与单质碳一样, 能够将黑色的氧化铜还原为红色的金属铜, 同时生成二氧化碳, 如图 2.1- 3 所示。

Similar to elemental carbon, carbon monoxide can reduce black copper oxide to red metallic copper, while producing carbon dioxide, as shown in Figure 2.1-3.

$$CuO + CO \xlongequal{\triangle} Cu + CO_2$$

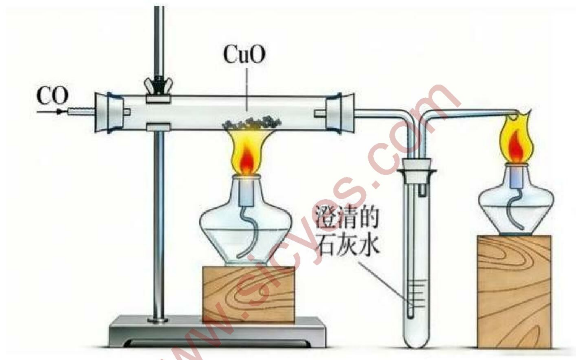

图 2.1-3 一氧化碳还原氧化铜

Figure 2.1-3 Carbon Monoxide Reducing Copper Oxide

我们可以利用一氧化碳的还原性，通过还原金属氧化物来制取某些金属。例如, 炼铁就是利用一氧化碳在高温下还原铁矿石 (ironstone) 里的氧化铁，同时生成二氧化碳。这个反应可以表示为:

We can utilize the reducing property of carbon monoxide to obtain certain metals by reducing metal oxides. For example, iron smelting is to use carbon monoxide to reduce iron oxide in iron ore at high temperatures, while producing carbon dioxide. This reaction can be expressed as:

$$Fe_2O_3 + 3CO \xlongequal{\text{高温/High temperature}} 2Fe + 3CO_2$$

3. 毒性

3. Toxicity

二氧化碳没有毒性(toxicity)，但一氧化碳有剧毒，原因是它能与血液中携带氧气的血红蛋白结合，破坏血液的输氧功能。因此，我们在生活中使用炉子、煤气时要格外小心，防止一氧化碳中毒。

Carbon dioxide is not toxic, but carbon monoxide is highly toxic because it can bind to hemoglobin in the blood that carries oxygen, destroying the oxygen-carrying function of the blood. Therefore, we should be extra careful when using stoves and gas in daily life to prevent carbon monoxide poisoning.

【例 3】一氧化碳在高温下还原铁矿石里的氧化铁，生成____和____。

【Example Problem 3】Carbon monoxide reduces iron oxide in iron ore at high temperatures, producing____ and____.

解题思路:首先明确该反应的反应物为氧化铁 (Fe_2O_3) 和一氧化碳 (CO)，反应条件为高温。根据一氧化碳的化学性质——还原性，它在高温下能夺取金属氧化物中的氧元素，将氧化铁中的铁元素还原为金属铁(Fe)，而一氧化碳自身会被氧化为二氧化碳(CO2)。结合给出的化学方程式，可直接判断出生成物为铁和二氧化碳。

Solution idea: First, clarify that the reactants of this reaction are iron oxide (Fe_2O_3) and carbon monoxide (CO) , and the reaction condition is high temperature. According to the chemical property of carbon monoxide - reducing property, it can 夺取 the oxygen element in the metal oxide at high temperatures, reducing the iron element in iron oxide to metallic iron (Fe), while carbon monoxide itself will be oxidized to carbon dioxide (CO2). Combining with the given chemical equation, it can be directly judged that the products are iron and carbon dioxide.

第二节 硅

Section 2 Silicon

硅(silicon, Si)是自然界中分布很广的一种元素。在地壳中，它的含量仅次于氧, 居第二位。自然界中没有游离态的硅, 只有以化合态存在的硅, 如二氧化硅、硅酸盐等。

Silicon (Si) is an element widely distributed in nature. In the earth's crust, its content ranks second only to oxygen. There is no free silicon in nature, only silicon in the form of compounds, such as silicon dioxide, silicates, etc.

一、单质硅

I. Elemental Silicon

1. 硅的物理性质

1. Physical Properties of Silicon

晶体硅是灰黑色、有金属光泽、硬而脆的固体, 它的结构类似于金刚石, 熔点和沸点都很高, 硬度也很大。晶体硅的导电性介于导体与绝缘体之间，是应用最为广泛的半导体材料。

Crystalline silicon is a grayish-black solid with a metallic luster, hard and brittle. Its structure is similar to that of diamond, with high melting and boiling points and great hardness. The conductivity of crystalline silicon is between that of conductors and insulators, and it is the most widely used semiconductor material.

2. 硅的化学性质

2. Chemical Properties of Silicon

硅的许多化学性质与碳相似。在常温下, 硅的化学性质不活泼, 除氟气、氢氟酸(hydrofluoric acid, HF)和强碱(strong alkali)外，硅不与其他物质, 如氧气、氯气、硫酸、硝酸等发生反应。在加热条件下，硅能与一些非金属发生反应。例如，加热时，硅粉能在氧气中燃烧，生成二氧化硅并放出大量的热。

Many chemical properties of silicon are similar to those of carbon. At room temperature, the chemical properties of silicon are not active. Except for fluorine gas, hydrofluoric acid (HF), and strong alkalis, silicon does not react with other substances such as oxygen, chlorine, sulfuric acid, nitric acid, etc. Under heating conditions, silicon can react with some non-metals. For example, when heated, silicon powder can burn in oxygen to form silicon dioxide and release a large amount of heat.

$$Si + O_2 \xlongequal{\triangle} SiO_2$$

自然界中没有单质硅存在, 因此我们使用的硅都是从它的化合物中提取出来的。在工业上, 利用碳在高温下还原二氧化硅的方法可制取含有少量杂质的粗硅。将粗硅提纯后，可以得到用作半导体材料的高纯硅。

There is no elemental silicon in nature, so the silicon we use is extracted from its compounds. Industrially, crude silicon containing a small amount of impurities can be produced by reducing silicon dioxide with carbon at high temperature. After purifying the crude silicon, high-purity silicon used as a semiconductor material can be obtained.

$$SiO_2 + 2C \xlongequal{\text{高温/High temperature}} Si + 2CO\uparrow$$

二、二氧化硅

II. Silicon Dioxide

二氧化硅(silicon dioxide, SiO_2)广泛存在于自然界，它的化学性质不活泼，不与水反应，也不与常见的酸(氢氟酸除外)反应，但能够与碱性氧化物或强碱反应生成盐。例如:

Silicon dioxide (silicon dioxide, SiO_2) exists widely in nature. Its chemical properties are not active, it does not react with water, nor with common acids (except hydrofluoric acid), but it can react with basic oxides or strong alkalis to form salts. For example:

$SiO_2 + 4HF = SiF_4\uparrow + 2H_2O$ (该反应可以用于刻蚀玻璃)

$SiO_2 + 4HF = SiF_4\uparrow + 2H_2O$ (This reaction can be used to etch glass)

$$SiO_2 + CaO \xlongequal{\text{高温/High temperature}} CaSiO_3$$

$$SiO_2 + 2NaOH = Na_2SiO_3 + H_2O$$

二氧化硅可用来生产光导纤维。二氧化硅是酸性氧化物，它对应的水化物是硅酸(silicic acid, H_2SiO_3)。硅酸不能直接由二氧化硅与水反应制得,只能通过可溶性硅酸盐与酸反应制取。硅酸不溶于水，是一种弱酸，酸性比碳酸还要弱。

Silicon dioxide can be used to produce optical fibers. Silicon dioxide is an acidic oxide, and its corresponding hydrate is silicic acid (silicic acid, H_2SiO_3). Silicic acid cannot be directly prepared by reacting silicon dioxide with water, but can only be prepared by reacting soluble silicates with acids. Silicic acid is insoluble in water, is a weak acid, and its acidity is weaker than that of carbonic acid.

$$Na_2SiO_3 + 2HCl = 2NaCl + H_2SiO_3\downarrow\ \text{(白色沉淀)}$$

三、硅酸盐

III. Silicates

还有一类硅化合物是硅酸盐(silicate)。硅酸盐是构成地壳岩石的主要成分, 自然界中存在的各种天然硅酸盐矿物的质量约占地壳质量的 5%。硅酸盐的种类很多, 结构也很复杂, 通常可用二氧化硅和金属氧化物的形式来表示其组成。例如:

There is also a class of silicon compounds called silicates (silicate). Silicates are the main components of the earth's crust rocks. The mass of various natural silicate minerals existing in nature accounts for about 5% of the earth's crust mass. There are many types of silicates and their structures are also very complex. Usually, their compositions can be expressed in the form of silicon dioxide and metal oxides. For example:

$$\text{硅酸钠}\ Na_2SiO_3(Na_2O \cdot SiO_2)$$

传统的无机非金属材料多为硅酸盐材料, 这类材料在日常生活中随处可见, 如制作餐具的陶瓷、窗户上的玻璃、建筑房屋用的水泥等。

Traditional inorganic non-metallic materials are mostly silicate materials. Such materials can be seen everywhere in daily life, such as ceramics for making tableware, glass on windows, cement for building houses, etc.

本章小结

Summary of This Chapter

1. 碳及其重要化合物的性质

1. Properties of Carbon and Its Important Compounds

(1)碳的化学性质

(1)Chemical Properties of Carbon

(2)二氧化碳的性质与鉴别

Properties and Identification of Carbon Dioxide

(3)二氧化碳的实验室制法

Laboratory Preparation of Carbon Dioxide

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Chemical Properties of Carbon Monoxide

2. 硅及其重要化合物的性质

2. Properties of Silicon and Its Important Compounds

(1)硅的化学性质

Chemical Properties of Silicon

(2)二氧化硅、硅酸及硅酸盐的性质

Properties of Silicon Dioxide, Silicic Acid and Silicates

(3)硅酸盐材料的应用

Applications of Silicate Materials

3. 涉及的方程式

3. Equations Involved

(1) $C + O_2 \overset{点燃/Ignition}{=\!=} CO_2$

(2) $2C + O_2 \overset{点燃/Ignition}{=\!=} 2CO$

(3) $2CuO + C \overset{高温/High\ temperature}{=\!=} 2Cu + CO_2\uparrow$

(4) $CO_2 + C \overset{高温/High\ temperature}{=\!=} 2CO$

(5) $CO_2 + H_2O = H_2CO_3$

(6) $H_2CO_3 = CO_2\uparrow + H_2O$

(7) $CO_2 + Ca(OH)_2 = CaCO_3\downarrow + H_2O$

(8) $CaCO_3 + 2HCl = CaCl_2 + CO_2\uparrow + H_2O$

(9) $2CO + O_2 \overset{点燃/Ignition}{=\!=} 2CO_2$

(10) $SiO_2 + 4HF = SiF_4\uparrow + 2H_2O$

(11) $Na_2SiO_3 + 2HCl = 2NaCl + H_2SiO_3\downarrow$ (白色沉淀/ White Precipitate)